PADMESH NAIK

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SUMMARY

Senior Software Engineer with 5+ years of experience building scalable applications, data pipelines, and AI/ML solutions. Experienced in Python, cloud technologies, MLOps, and Generative AI, with a strong track record of delivering high-impact, production-grade systems and end-to-end data solutions.

SKILLS

Languages: Python, SQL, Java, JavaScript
AI Engineering: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain
Data & ML: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Apache Spark, Kafka
Databases & Backend: Oracle, PostgreSQL, MySQL, MongoDB, Django, REST APIs, Pinecone
Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Jenkins, Airflow, Git, Unit Testing, Mocking

EDUCATION

Master of Science in Computer Science | Worcester Polytechnic Institute (WPI), MA **May 2024**
Bachelors in Computer Engineering | University of Mumbai, India **October 2020**

PROFESSIONAL EXPERIENCE

Python Developer | Bank of America (Contract), New Jersey **February 2025 – Present**

- Enhanced Basel III credit risk models in **Python**, updating Probability of Default (PD), Loss Given at Default (LGD), and Exposure at Default (EAD) calculations and correcting Risk-Weighted Assets (RWA) by approximately **$2.1 billion.**
- Built end-to-end Python and **Oracle** data pipeline enhancements for new credit-risk parameters and regulatory fields, covering ingestion, transformation, calculation, and reporting.
- Fixed an exposure-calculation defect that understated credit risk by approximately **$400 million**, thus improving the accuracy of Risk-Weighted Assets (RWA) and regulatory capital calculations.
- Automated **70+ SQL**-based smoke-test validations using Python, reducing validation effort by **~90%** from hours to minutes while achieving 100% table coverage.
- Delivered and supported production releases across System Integration Testing (SIT), User Acceptance Testing (UAT), Pre-Production (PREPROD), and Production (PROD), including **unit** and **regression testing**, peer reviews, debugging, deployment, and post-release validation.
- Resolved production data-pipeline issues including Cartesian joins, data-type mismatches, failed file archival, and **environment migration inconsistencies**, reducing affected job runtime and improving pipeline reliability.

Software Engineer | CTP, New York **August 2024 – January 2025**

- Created a scalable e-commerce platform using **Shopify** themes, Liquid templating, and **JavaScript**, integrating custom plugins (SEO Booster) to enable dynamic product filtering, personalized recommendations, and automated order tracking functionality.
- Developed and deployed backend infrastructure using **AWS** EC2, RDS, API Gateway, VPC, and Secrets Manager, implementing secure networking and automated database operations to support global accessibility and concurrent user requests.
- Developed **REST APIs** and optimized **MySQL** database schemas with efficient indexing strategies, utilizing AWS infrastructure to enhance query performance and enable seamless integration between Shopify frontend and backend services.

Software Developer Intern | Clearly Energy, Maryland **January 2024 – April 2024**

- Engineered a web application using **Angular.js** and **Django**, implementing interactive maps and energy policy compliance tracking features for building management across multiple jurisdictions.
- Implemented **Docker** containerization and **Jenkins CI/CD** pipelines for automated deployment and testing, reducing deployment failures by **18%** and enhancing application reliability.
- Developed comprehensive geospatial features using **OpenLayers**, including address search functionality, dynamic filter menus, and data points clustering for efficient visualization.
- Optimized database performance through schema normalization and SQL indexing strategies, resulting in **22%** improvement in data retrieval efficiency for large-scale property datasets.

Software Engineer Intern | Building Assure PBC, Massachusetts **May 2023 – August 2023**

- Implemented a real-time IoT monitoring system using **React** and **Node.js**, integrating GoRules Engine for decision-making, resulting in streamlined alert generation and enhanced system responsiveness for environmental metrics including CO2, temperature, and humidity.
- Engineered scalable data pipelines using **AWS Lambda, Python,** and **Apache Kafka** for real-time data streaming and processing, integrated with MongoDB for efficient data persistence, leading to a 25% improvement in data processing speed and enhanced system scalability.
- Developed comprehensive API endpoints and **MongoDB** database operations for IoT sensor data management, implementing rule-based alert generation systems and real-time data transformation logic across multiple environmental parameters.
- Established robust testing infrastructure using **Jest** and **Mocha**, creating unit and acceptance tests for API endpoints and product features, ensuring system reliability and maintaining code quality through automated test coverage.

System Software Engineer | Tata Consultancy Services, India **September 2020 - August 2022**

- Architected and implemented end-to-end CI/CD pipelines for microservices-based absenteeism analysis system using **Jenkins, Docker,** and **Kubernetes**, reducing deployment time by **30%** and improving release reliability through automated testing and containerization.
- Developed and optimized ETL data pipelines for processing employee records using **Python, Apache Spark,** and **Snowflake**, reducing processing time from **9 hours to 3.5 hours** by implementing parallel processing, enabling timely workforce planning and reporting.
- Engineered machine learning models (**XGBoost** and **LightGBM**) for estimating time of arrival (ETA) in the automotive parts supply chain, integrating AWS S3, Snowflake and TensorFlow to enhance delivery accuracy and optimize inventory management.
- Collaborated with product teams to translate business requirements into technical specifications, conducted root cause analysis using logging and monitoring tools, and implemented targeted fixes, achieving **91% user satisfaction** in post-release surveys.
- Led data analysis initiatives using Python and SQL to analyze 30 years of customer purchase patterns, implementing **Random Forest** models for churn prediction and price optimization, improving customer retention through targeted strategies.

PROJECTS

RoleReady AI: Developed an **LLM-powered** interview application using Python, **LangChain**, OpenAI **GPT-4**, Pinecone, and Streamlit, generating role-specific questions and automated scoring/feedback. Implemented **Retrieval-Augmented Generation (RAG)** with vector embeddings and **Pinecone** similarity search, and optimized API costs through prompt engineering, conversation summarization, caching, and a separate lower-cost scoring model.

MLOps for Sales Prediction: Built an end-to-end machine learning pipeline using Python, Jenkins, Docker, **Apache Airflow, MLflow**, and AWS, automating data ingestion, model training, evaluation, deployment, and monitoring for sales forecasting. Implemented **CI/CD** and experiment tracking to create a reproducible and scalable machine-learning workflow.

PUBLICATIONS

Image Forgery: Detection of Manipulated Images Using Neural Network:

Image Forgery: Detection of Manipulated Images Using Neural Network — Developed a CNN-based image manipulation detection approach using Error Level Analysis; featured on SSRN's **Top 10 download list** with 600+ downloads.

CERTIFICATIONS

AWS Certified Cloud Practitioner
Azure Fundamentals
Azure Data Scientist
Generative AI with LLMs